

Triangles, radians and sectors - Set 1 - Solutions

Specification MM4.1, MM4.2, M5.16.

Answer key

Question	Answer	Topic
1	F	Triangles, radians and sectors
2	B	Triangles, radians and sectors
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Worked solutions

Question 1 TRI-01 · Triangles, radians and sectors · spec MM4.1

In triangle ABC , $AB = 5$, $AC = 8$ and angle $BAC = 60^\circ$. Find BC .

Answer: F

Cosine rule: $BC^2 = 25 + 64 - 2 \cdot 5 \cdot 8 \cdot \frac{1}{2} = 89 - 40 = 49$, so $BC = 7$.

Full method - how and why

Two sides and the angle between them: that is exactly the cosine rule's input.

- 1 Choose the rule: you know two sides and the **included** angle, so use the cosine rule, not the sine rule.

why: The sine rule needs a side opposite a known angle; here the known angle sits between the known sides.

- 2 Write it with BC opposite the known angle A : $BC^2 = AB^2 + AC^2 - 2(AB)(AC)\cos A$.

why: The side being found must be opposite the angle being used.

- 3 Substitute, using $\cos 60^\circ = \frac{1}{2}$: $BC^2 = 25 + 64 - 2(5)(8)(\frac{1}{2}) = 89 - 40 = 49$.

why: $\cos 60^\circ = \frac{1}{2}$ is one of the exact values you must recall.

- 4 Take the square root: $BC = 7$.

why: Lengths are positive, so only the positive root.

Where it goes wrong

- Forgetting the $-2bc\cos A$ term and answering $\sqrt{89}$ - which is offered.
- Using $\cos 60^\circ = \frac{\sqrt{3}}{2}$ (that is $\cos 30^\circ$), giving $\sqrt{89 - 40\sqrt{3}}$.

Takeaway Two sides plus the included angle $\Rightarrow$ cosine rule; the unknown side is opposite the known angle.

Question 2 TRI-02 · Triangles, radians and sectors · spec MM4.1

A triangle has sides 4, 5 and 6. Find the cosine of its largest angle.

Answer: B

The largest angle is opposite the side 6: $\cos \theta = \frac{16 + 25 - 36}{2 \cdot 4 \cdot 5} = \frac{5}{40} = \frac{1}{8}$.

Full method - how and why

Three sides given, an angle wanted: the rearranged cosine rule. The word 'largest' tells you which angle.

- 1 Identify the largest angle: it is opposite the longest side, 6.

why: In any triangle the ordering of sides matches the ordering of opposite angles.

- 2 Use $\cos \theta = \frac{a^2 + b^2 - c^2}{2ab}$ with $a = 4$, $b = 5$, $c = 6$.

why: The two sides adjacent to the angle go on the bottom; the opposite side is the one subtracted on top.

- 3 Substitute: $\frac{16 + 25 - 36}{2(4)(5)} = \frac{5}{40} = \frac{1}{8}$.

why: Positive, so the largest angle is acute - this triangle has no obtuse angle at all.

Where it goes wrong

- Putting the longest side on the bottom, giving $\frac{9}{16}$ or similar.
- Assuming the largest angle must be obtuse and choosing $-\frac{1}{8}$.

Takeaway $\cos \theta = \frac{a^2 + b^2 - c^2}{2ab}$ with c opposite θ ; a positive result means an acute angle.

Question 3 TRI-03 · Triangles, radians and sectors · spec MM4.2

A sector of a circle of radius 6 cm has area $15\pi \text{ cm}^2$. Find the arc length of the sector.

Answer: C

$$\frac{1}{2}r^2\theta = 18\theta = 15\pi \Rightarrow \theta = \frac{5\pi}{6}. \text{ Arc} = r\theta = 6 \cdot \frac{5\pi}{6} = 5\pi.$$

Full method - how and why

Sector area gives the angle; the angle gives the arc length. Both formulas need radians.

- 1 Use the sector area formula $A = \frac{1}{2}r^2\theta$: $\frac{1}{2}(36)\theta = 18\theta$.
why: $r = 6$, so $\frac{1}{2}r^2 = 18$. The formula is only valid with θ in radians.
- 2 Set equal to the given area: $18\theta = 15\pi$, so $\theta = \frac{5\pi}{6}$.
why: Divide: $\frac{15\pi}{18} = \frac{5\pi}{6}$.
- 3 Use the arc-length formula $s = r\theta = 6 \times \frac{5\pi}{6} = 5\pi$ cm.
why: The 6 cancels neatly - a sign the angle is right.

Where it goes wrong

- Quoting the angle $\frac{5\pi}{6}$ as the arc length - it is offered as a distractor.
- Using $A = r^2\theta$ without the $\frac{1}{2}$, which halves the angle and gives $\frac{5\pi}{2}$.

Takeaway $A = \frac{1}{2}r^2\theta$ and $s = r\theta$ (radians); find θ from one, then substitute into the other.

Question 4 TRI-05 · Triangles, radians and sectors · spec MM4.2

Convert 135° to radians.

Answer: C

$$135 = \frac{3}{4} \times 180, \text{ so } \frac{3\pi}{4}.$$

Full method - how and why

Degree-to-radian conversion: multiply by $\frac{\pi}{180}$, or spot the fraction of 180° .

- 1 Express 135 as a fraction of 180: $\frac{135}{180} = \frac{3}{4}$.
why: Since $180^\circ = \pi$ radians, the same fraction of π gives the answer.
- 2 So $135^\circ = \frac{3}{4}\pi = \frac{3\pi}{4}$.
why: Equivalently $135 \times \frac{\pi}{180} = \frac{3\pi}{4}$.

Where it goes wrong

- Multiplying by $\frac{180}{\pi}$ (the reverse conversion).
- Simplifying $\frac{135}{180}$ incorrectly to $\frac{2}{3}$ or $\frac{5}{6}$, both of which are offered.

Takeaway $180^\circ = \pi$: convert by multiplying by $\frac{\pi}{180}$, and cancel the fraction fully.

Question 5 TRI-09 · Triangles, radians and sectors · spec MM4.1

Two sides of a triangle are 3 and 5 and the included angle is θ , where $\cos \theta = -\frac{1}{5}$. Find the length of the third side.

Answer: C

$$c^2 = 9 + 25 - 2 \cdot 3 \cdot 5 \cdot \left(-\frac{1}{5}\right) = 34 + 6 = 40.$$

Full method - how and why

Cosine rule with a negative cosine. The only difficulty is handling the double negative.

- 1 Write the rule: $c^2 = 3^2 + 5^2 - 2(3)(5)\cos \theta$.
why: Two sides and the included angle again.
- 2 Substitute $\cos \theta = -\frac{1}{5}$: $c^2 = 9 + 25 - 30\left(-\frac{1}{5}\right) = 34 + 6$.
why: Subtracting a negative adds: $-30 \times \left(-\frac{1}{5}\right) = +6$.
- 3 So $c^2 = 40$ and $c = \sqrt{40}$.
why: Leave it in surd form; $\sqrt{40} = 2\sqrt{10}$ if you prefer.

Where it goes wrong

- Getting $34 - 6 = 28$ by mishandling the sign, which is offered as $\sqrt{28}$.
- Forgetting the angle is obtuse, so the third side must be **longer** than $\sqrt{34}$ - a useful sanity check.

Takeaway A negative cosine makes the third side longer than Pythagoras would give; check your answer against $\sqrt{a^2 + b^2}$.

Question 6 TRI-13 · Triangles, radians and sectors · spec MM4.2

The minute hand of a clock is 12 cm long. How far does its tip travel in 25 minutes?

Answer: C

$$25 \text{ minutes is } \frac{25}{60} = \frac{5}{12} \text{ of a revolution: } \frac{5}{12} \times 2\pi \times 12 = 10\pi \text{ cm.}$$

Full method - how and why

Distance travelled by a rotating tip is arc length. Find the fraction of a full turn, then multiply by the circumference.

- 1 Find the fraction of a revolution: $\frac{25}{60} = \frac{5}{12}$.
why: The minute hand takes 60 minutes for one full turn.
- 2 Compute the full circumference: $2\pi r = 2\pi(12) = 24\pi$ cm.
why: The tip traces a circle of radius equal to the hand's length.
- 3 Multiply: $\frac{5}{12} \times 24\pi = 10\pi$ cm.
why: Equivalently, $\theta = \frac{5}{12} \times 2\pi = \frac{5\pi}{6}$ and $s = r\theta = 12 \times \frac{5\pi}{6} = 10\pi$.

Where it goes wrong

- Using the radius as the distance, or the area formula instead of arc length.
- Using $\frac{25}{12}$ or $\frac{25}{60}$ of the **radius** rather than the circumference.

Takeaway Arc length = (fraction of a turn) $\times 2\pi r$; a clock hand makes one turn per hour.

Question 7 X2-15 · Triangles, radians and sectors · spec MM4.1

In triangle ABC , P : ' $a^2 = b^2 + c^2$ '. Q : 'angle $A = 90^\circ$ '. Which one of the following is true?

Answer: A

Pythagoras and its converse: the cosine rule gives $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$, which is 0 exactly when $a^2 = b^2 + c^2$.

Full method - how and why

Pythagoras **and** its converse both hold, so this is a genuine equivalence - best seen through the cosine rule.

1 The cosine rule gives $a^2 = b^2 + c^2 - 2bc \cos A$, so $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$.

why: Rearranging the cosine rule turns both statements into one equation.

2 $a^2 = b^2 + c^2$ makes the numerator zero, so $\cos A = 0$ and (in a triangle) $A = 90^\circ$.

why: $2bc \neq 0$ since side lengths are positive; and $0^\circ < A < 180^\circ$ makes $A = 90^\circ$ the only solution of $\cos A = 0$.

3 Conversely $A = 90^\circ$ gives $\cos A = 0$, so $b^2 + c^2 - a^2 = 0$.

why: The same equation, read the other way.

4 So P and Q are equivalent: necessary and sufficient.

why: This is Pythagoras together with its converse.

Where it goes wrong

- Assuming only the forward direction (right angle $\Rightarrow$ Pythagoras) and answering 'necessary but not sufficient'.
- Forgetting that a must be the side **opposite** A for the relation to hold.

Takeaway The cosine rule contains Pythagoras and its converse: $\cos A = 0 \Leftrightarrow a^2 = b^2 + c^2$, with a opposite A .

Question 8 TRI-04 · Triangles, radians and sectors · spec MM4.1

Triangle PQR has area $10\sqrt{3}$, $PQ = 8$ and $PR = 5$. Given that angle QPR is obtuse, find QR .

Answer: D

$\frac{1}{2} \cdot 8 \cdot 5 \sin P = 10\sqrt{3} \Rightarrow \sin P = \frac{\sqrt{3}}{2}$, and P obtuse gives $P = 120^\circ$. $QR^2 = 64 + 25 - 80 \cos 120^\circ = 89 + 40 = 129$.

Full method - how and why

Area gives $\sin P$, which has two possible angles. The word 'obtuse' picks the right one, and then the cosine rule finishes.

1 Use area $= \frac{1}{2}ab \sin C$: $\frac{1}{2}(8)(5) \sin P = 10\sqrt{3}$, so $20 \sin P = 10\sqrt{3}$ and $\sin P = \frac{\sqrt{3}}{2}$.

why: The formula uses the angle **between** the two sides, which is QPR here.

2 $\sin P = \frac{\sqrt{3}}{2}$ gives $P = 60^\circ$ or $P = 120^\circ$; the obtuse condition selects $P = 120^\circ$.

why: $\sin \theta = \sin(180^\circ - \theta)$; a sine value never determines the angle by itself.

3 Apply the cosine rule with $\cos 120^\circ = -\frac{1}{2}$: $QR^2 = 64 + 25 - 2(8)(5)(-\frac{1}{2}) = 89 + 40 = 129$.

why: A negative cosine turns the subtraction into an addition, making the opposite side longer - which is what an obtuse angle should do.

4 So $QR = \sqrt{129}$.

why: Left in surd form; no calculator is available anyway.

Where it goes wrong

- Taking $P = 60^\circ$ and answering 7 - the value the question's 'obtuse' is designed to exclude.
- Using $\cos 120^\circ = +\frac{1}{2}$, giving $\sqrt{49}$.

Takeaway $\sin \theta$ always allows two angles; use any stated acute/obtuse condition, and remember $\cos$ is negative for obtuse angles.

Question 9 TRI-06 · Triangles, radians and sectors · spec MM4.2

A chord of a circle of radius 4 subtends an angle of $\frac{\pi}{3}$ at the centre. Find the area of the minor segment cut off by the chord.

Answer: E

Sector area $\frac{1}{2} \cdot 16 \cdot \frac{\pi}{3} = \frac{8\pi}{3}$. Triangle area $\frac{1}{2} \cdot 16 \sin \frac{\pi}{3} = 4\sqrt{3}$. Segment = $\frac{8\pi}{3} - 4\sqrt{3}$.

Full method - how and why

Segment = sector – triangle. Compute the two areas and subtract; both use the same r and θ .

1 Sector area: $\frac{1}{2}r^2\theta = \frac{1}{2}(16)(\frac{\pi}{3}) = \frac{8\pi}{3}$.

why: $r = 4$ so $r^2 = 16$; the angle is already in radians.

2 Triangle area (two radii and the chord): $\frac{1}{2}r^2 \sin \theta = \frac{1}{2}(16) \sin \frac{\pi}{3} = 8 \times \frac{\sqrt{3}}{2} = 4\sqrt{3}$.

why: Use $\frac{1}{2}ab \sin C$ with both sides equal to the radius.

3 Subtract: segment = $\frac{8\pi}{3} - 4\sqrt{3}$.

why: The minor segment is the bit of the sector left over once the triangle is removed.

Where it goes wrong

- Forgetting the $\frac{1}{2}$ in the triangle area, giving $\frac{8\pi}{3} - 8\sqrt{3}$ (an offered option), which is negative.
- Using $\sin \frac{\pi}{3} = \frac{1}{2}$.

Takeaway Segment area = $\frac{1}{2}r^2(\theta - \sin \theta)$ - sector minus triangle, all in radians.

Question 10 TRI-07 · Triangles, radians and sectors · spec MM4.1

In triangle ABC , $AB = 6$, $BC = 4$ and angle $BAC = 30^\circ$. How many possible triangles are there?

Answer: C

Sine rule: $\frac{\sin C}{6} = \frac{\sin 30^\circ}{4}$, so $\sin C = \frac{3}{4}$. Since $BC = 4 < AB = 6$ and $4 > 6 \sin 30^\circ = 3$, both the acute and obtuse values of C give valid triangles: the ambiguous case, two triangles.

Full method - how and why

Side-side-angle data: the ambiguous case. Compute $\sin C$, then decide whether both the acute and obtuse angles give a valid triangle.

1 Apply the sine rule: $\frac{\sin C}{AB} = \frac{\sin A}{BC}$, i.e. $\frac{\sin C}{6} = \frac{\sin 30^\circ}{4}$.

why: Pair each angle with the side opposite it: C is opposite $AB = 6$, and A is opposite $BC = 4$.

2 Solve: $\sin C = \frac{6 \times 0.5}{4} = \frac{3}{4}$, which is less than 1, so solutions exist.

why: If $\sin C$ came out greater than 1, no triangle would exist at all.

3 Both $C = \arcsin \frac{3}{4}$ and $180^\circ - \arcsin \frac{3}{4}$ are possible in principle; check whether the obtuse one still leaves a positive third angle.

why: $\arcsin \frac{3}{4} \approx 48.6^\circ$, so the obtuse option is about 131.4° ; with $A = 30^\circ$ that totals under 180° ✓.

4 Equivalently: the side opposite the given angle (4) is shorter than the other given side (6) but longer than $6 \sin 30^\circ = 3$ - the classic two-triangle condition. So there are 2 triangles.

why: The condition $b \sin A < a < b$ is the standard test for the ambiguous case.

Where it goes wrong

- Assuming one triangle because the sine rule 'gives one answer'.
- Assuming two triangles always - if $a \geq b$ the obtuse option would make the angles exceed 180° .

Takeaway SSA data: with $b \sin A < a < b$ there are two triangles; with $a \geq b$ exactly one; with $a < b \sin A$ none.

Question 11 TRI-08 · Triangles, radians and sectors · spec MM4.2

The perimeter of a sector of radius r and angle θ radians is 20, and its area is 16. Find the possible values of r .

Answer: D

$2r + r\theta = 20$ and $\frac{1}{2}r^2\theta = 16$. From the first $r\theta = 20 - 2r$; then $\frac{1}{2}r(20 - 2r) = 16 \Rightarrow r^2 - 10r + 16 = 0 \Rightarrow r = 2$ or 8 .

Full method - how and why

Two sector facts, two unknowns. Eliminate θ by substitution and solve the resulting quadratic in r .

- 1 Perimeter of a sector = two radii + arc: $2r + r\theta = 20$.

why: Students often forget the two straight edges - the perimeter is not just the arc.

- 2 Rearrange for the arc: $r\theta = 20 - 2r$.

why: Keeping the product $r\theta$ together is the key, because the area formula contains it.

- 3 Area: $\frac{1}{2}r^2\theta = \frac{1}{2}r(r\theta) = \frac{1}{2}r(20 - 2r) = 16$.

why: Writing $r^2\theta$ as $r \times (r\theta)$ lets you substitute directly - no need to find θ .

- 4 Simplify and solve: $10r - r^2 = 16 \Rightarrow r^2 - 10r + 16 = 0 \Rightarrow (r - 2)(r - 8) = 0$, so $r = 2$ or 8 .

why: Both are geometrically valid: $r = 2$ gives $\theta = 8$ radians... which exceeds 2π , but the question asks only for the values of r from the equations.

Where it goes wrong

- Using perimeter = $r\theta$ alone, which gives a different quadratic.
- Finding only one root of the quadratic.

Takeaway Substitute the product $r\theta$ as a block; sector perimeter is $2r + r\theta$.

Question 12 TRI-10 · Triangles, radians and sectors · spec MM4.1

In triangle ABC , $\sin A : \sin B : \sin C = 3 : 5 : 7$. Find angle C .

Answer: A

By the sine rule the sides are in ratio $3 : 5 : 7$. $\cos C = \frac{9 + 25 - 49}{2 \cdot 3 \cdot 5} = -\frac{15}{30} = -\frac{1}{2}$, so $C = 120^\circ$.

Full method - how and why

A ratio of sines is a ratio of sides (sine rule). Then the cosine rule finds the angle.

- 1 By the sine rule, $a : b : c = \sin A : \sin B : \sin C = 3 : 5 : 7$.

why: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$ means sides are proportional to the sines of opposite angles.

- 2 Take the sides as $3k, 5k, 7k$; the k will cancel, so use $3, 5, 7$.

why: Only the shape matters for angles, so the scale factor is irrelevant.

- 3 Apply the cosine rule for the angle opposite 7: $\cos C = \frac{9 + 25 - 49}{2(3)(5)} = \frac{-15}{30} = -\frac{1}{2}$.

why: C is opposite the largest side, so expect the largest angle.

- 4 $\cos C = -\frac{1}{2}$ gives $C = 120^\circ$.

why: One of the exact values: $\cos 120^\circ = -\cos 60^\circ = -\frac{1}{2}$.

Where it goes wrong

- Treating $3 : 5 : 7$ as the angles themselves, giving 84° .
- Reading $\cos C = -\frac{1}{2}$ as 60° and ignoring the sign.

Takeaway Sines in ratio $\Rightarrow$ sides in the same ratio; then use the cosine rule with any convenient multiple.